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Full Length Article

Modelling fuel flexibility in fixed-bed biomass conversion with a low primary air ratio in an updraft configuration

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ABSTRACT

Fixed-bed biomass conversion with a low primary air ratio and a counter-current configuration has a high feedstock flexibility, as it resembles updraft gasification, and the potential to reduce emissions when integrated in biomass combustion systems. A 1D bed model was validated with experimental results from a biomass combustion boiler with such a bed conversion system, predicting with a good accuracy the temperatures in the reactor and producer gas composition. The model was applied for different cases to investigate the fuel flexibility of this combustion system, including the influence of moisture content and the maximum temperatures achieved in the bed. It was shown that with variations in fuel moisture content from 8 to 30% mass w.b. the producer gas composition, char reduction to CO or maximum temperatures at the grate were not affected due to the separation of the char conversion and pyrolysis/drying zones. Flue gas recirculation was the only possible measure with the tested configuration to reduce the maximum temperatures close to the grate, which is beneficial e.g. to avoid slagging with complicated fuels. A higher tar content was obtained than in conventional updraft gasifiers, which is attributed to the absence of tar condensation in the bed due to the limited height of the reactor and the integration in the combustion chamber. The presented model can support the development of such combustion technologies and is a relevant basis for detailed CFD simulations of the bed or gas phase conversion.

1. Introduction

Bioenergy is the most relevant renewable energy source nowadays and the most common use is bioheat obtained from combustion of lignocellulosic biomass [1]. This is a well-known process and the emissions of unburnt products as CO or soot in modern and automatic conversion devices with air staging are generally low. However, still significant emissions are produced of nitrogen oxides (NO_X) and inorganic particle matter (PM) in these devices [2,3], and the feedstock flexibility is limited, especially in small scales [1,4].

The bed conversion in combustion with air staging takes place at under-stoichiometric conditions, commonly with a primary excess air ratio (λ) between 0.6 and 1 [1,5]. Recent developments go in the direction of employing lower primary air excess ratios. In this way bed temperatures are reduced, increasing the feedstock flexibility and reducing the release of ash-forming elements which leads to PM emissions [6–8]. Reducing λ in the fuel bed to values lower than 0.4 in a fixed-bed counter-current configuration leads to conversion in an oxygen limited regime, significantly increasing the thickness of the char conversion zone [9]. Fixed-bed combustion with a low oxygen concentration resembles then the configuration of an updraft gasifier [6]. Conventional updraft gasification can be conducted with a simple geometry and it has a high feedstock flexibility, regarding particle size or moisture content which can be up to 60% mass w.b. [10,11]. In updraft gasification there is as well a high retention of potassium in the bed [12], leading to a low dust content. These characteristics are positive to overcome current main limitations in fixed-bed biomass combustion, as high inorganic PM emissions and, in some cases, limited feedstock flexibility. The main disadvantage of updraft gasification is the high tar content in the producer gas, which limits the possible uses of the producer gas, but it is not relevant if the producer gas is directly burnt above the fixed-bed.

The main features of conversion in countercurrent or updraft gasification with air were described with one-dimensional models for biomass by Di Blasi [13] and for coal by Hobbs et al. [14]. Other models

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Nomenclature

A	Pre-exponential factor, $s^{-1}/m\cdot s^{-1}$
A_p	Speci
c	particle surface area, $m^2\cdot m^{-3}$
c	Gas concentration, $mol\cdot m^{-3}$
c_p	Heat capacity, $J\cdot kg^{-1}\cdot K^{-1}$
d	Diameter, m
E	Activation energy, $J\cdot mol^{-1}$
Mm	Molecular weight, $kg\cdot mol^{-1}$
p	Pressure, Pa
Q	Heat transfer rate, $W\cdot m^{-3}$
R	Gas constant, $J\cdot mol^{-1}\cdot K^{-1}$
$\dot{r}$	Reaction rate, $kg\cdot m^{-3}\cdot s^{-1}$
T	Temperature, K
t	Time, s
V	Volumen, m^3
v	Superficial velocity, $m\cdot s^{-1}$
X	Mol fraction/Conversion, –
Y	Mass fraction, –
z	Axial coordinate, m
α	Mols H/mol C in char, –
β	Mass transfer coefficient/Mols O/mol C in char, $m\cdot s^{-1}$ /–
ε	Porosity, –
η	CO/CO_2 ratio, –
κ	Permeability, m^2
λ	Thermal conducvitiy, $W\cdot m^{-1}\cdot K^{-1}$
μ	Viscosity, $kg\cdot m^{-1}\cdot s^{-1}$
ρ	Density, $kg\cdot m^{-3}$
Δh	Enthalpy of combustion/reaction, $J\cdot kg^{-1}$
Subscript	
g	Gas,
i	Species “i”
k	Reaction “k”
l	Lateral
p	Particle
r	Reactor
s	Solid
0	Initial

have been presented in literature for gasification of wood pellets [15], gasification with high temperature air [16], gasification with high temperature steam [17], or steam plasma gasification of municipal solid waste [18]. Models with similar features, besides the boundary conditions, are as well available for fixed-bed downdraft gasifiers [19,20]. These models are quasi-continuous models for the solid and gas phases, employing simplified pyrolysis models. A detailed pyrolysis model was employed in [21], coupled with a particle model at each layer of an updraft reactor model, describing the reactor with 10 layers. The previous models were validated with comparisons to measurements of temperature profiles along the reactor height and producer gas composition of a limited number of cases. Detailed experimental data from updraft gasifiers in literature is however scarce, and only a few works presented the producer gas composition experimentally measured along the reactor height [22,23].

The objective of this work is to develop a state-of-the-art updraft gasification model with similar features as the previously presented models and to apply it to describe the bed conversion in a counter-current combustion system with a low primary air ratio in the fixed-bed. Experimental data from such a system was recently presented by this group [6] and will be employed for model validation. The focus and novelty of this work will be to investigate the influence of fuel moisture content and the achieved maximum temperatures in the grate to gain further insights in the benefits regarding feedstock flexibility that can be obtained for biomass combustion with such systems. This will support the development of novel biomass combustion systems which minimize PM emissions in the flue gas while having a high fuel flexibility.

2. Model

The presented model is a one-dimensional quasi-continuous volumetric model which has the following features and assumptions:

- Two phases are considered: the solid (and liquid) phase with 3 species (dry biomass, char and moisture), and the gas phase with 9 species (CO, CO_2, H_2O, H_2, O_2, CH_4, C_2H_4, tar and N_2).
- The species mass transport equations and energy equations are solved along the reactor height. The transport equations for gas and solid species are shown in Equation (1) and Equation (2), respectively, and the energy equations for the gas and solid phases are shown in Equation (3) and Equation (4), respectively. The energy equations are presented in a non-conservative form in order to directly calculate the temperatures and are valid for low Mach number fluids [24].
- The considered reactions are shown in Table 1 and include drying, pyrolysis and char conversion. Char conversion is described with the shrinking core model as it will be detailed later on.
- The gas velocities are calculated through the Darcy law shown in Equation (5); while pressure, gas temperature and density are coupled through the equation of state, shown in Equation (6).
- The required boundary conditions are the biomass flow and composition, including moisture content, the inlet gas flow, composition and temperature, thermal resistance of the insulation and outside temperature, heat loss through the bottom (grate) and heat input from the top (from radiation of the combustion chamber). Atmospheric pressure is assumed at the reactor outlet and for other scalar variables, Neumann conditions are considered.

$$\frac{\partial(\varepsilon_r\rho_g Y_i)}{\partial t} = -\frac{\partial(\rho_g Y_i v_g)}{\partial z} + \dot{r}_i \tag{1}$$

$$\frac{\partial((1-\varepsilon_r)\rho_s Y_i)}{\partial t} = -\frac{\partial(\rho_s Y_i v_s)}{\partial z} + \dot{r}_i \tag{2}$$

$$\varepsilon_r\rho_g c_{p,g}\frac{\partial T_g}{\partial t} = -\rho_g c_{p,g} v_g\frac{\partial T_g}{\partial z} + \frac{\partial}{\partial z}\left(\lambda_g\frac{\partial T_g}{\partial z}\right) + \dot{Q}_{s,g} - \dot{Q}_{g,l} \tag{3}$$

$$(1-\varepsilon_r)\rho_s c_{p,s}\frac{\partial T_s}{\partial t} = -\rho_s c_{p,s} v_s\frac{\partial T_s}{\partial z} + \frac{\partial}{\partial z}\left(\lambda_s\frac{\partial T_s}{\partial z}\right) - \sum_k \dot{r}_k\Delta h_k - \dot{Q}_{s,g} - \dot{Q}_{s,l} \tag{4}$$

$$v_g = -\frac{\kappa_s}{\mu_g}\frac{\partial p}{\partial z} \tag{5}$$

$$\rho_g = -\frac{pM_m}{RT_g} \tag{6}$$

Drying and pyrolysis are described with an Arrhenius equation, as shown in Equation (7), being “i“ moisture or dry biomass, respectively. The particle size and solid velocity are not modified during these reactions, while intra-particle gradients are neglected. Only changes in the bulk density will take place due to conversion. The kinetics [15,25] and product composition are shown in Table 1. The composition of the pyrolysis products is derived from the application last version of the detailed RAC scheme [26,27], which is especially suited for pyrolysis in a fixed-bed where secondary charring reactions are of high relevance.

Table 1
Reactions.

Reaction	Stoichiometry (mass fraction for pyrolysis, mol fraction others)	A	E ($kJ\cdot mol^{-1}$)	Source
Drying	$H_2O(l) \rightarrow H_2O(g)$	5.6×10^6 s^{-1}	88	[15]
Pyrolysis	Biomass → 0.234 Char ($CH_\alpha O_\beta$, α = 0.3934, β = 0.0484) + 0.082 CO + 0.114 H_2O + 0.124 CO_2 + 0.006 H_2 + 0.016 CH_4 + 0.013 C_2H_4 + 0.411 Tar ($C_{2.3466}H_{3.9671}O_{1.5296}$)	2×10^8 s^{-1}	133	Kinetics: [25]Composition: [26,27]
Char oxidation	Char ($CH_\alpha O_\beta$) + (1 - η/2 + α/4 - β/2) O_2 → η CO + (1-η) CO_2 + α/2 H_2O	5.7×10^7 $m\cdot s^{-1}$	160	Kinetics: [13]η: [30] – see Equation (11)
CO_2 char gasification	Char ($CH_\alpha O_\beta$) + CO_2 → 2 CO + (α/2 - β) H_2 + β H_2O	1×10^7 $m\cdot s^{-1}$	220	[13]
Steam char gasification	Char ($CH_\alpha O_\beta$) + (1 – β) H_2O → CO + (1 + α/2 - β) H_2	1×10^7 $m\cdot s^{-1}$	220	[13]

This scheme is applied for a typical softwood composition (44% cellulose, 26% hemicellulose, 30% lignin – 17.5% LIG-C, 9.5% LIG-H and 3% LIG-O) [28], at a slow heating rate (20 K/min) and until a temperature of 700 °C (at higher temperatures char conversion reactions are active [29]). High charring conditions are applied, which are typical for slow conversion in a fixed-bed (see details in [26,27]). The condensable and solid products obtained as final yields from the RAC scheme are grouped respectively in the single tar and char species. Both tar and char compositions include carbon, oxygen and hydrogen, and they are detailed in Table 1.

$$\dot{r}_{drying/pyrolysis} = \rho_s Y_i (1-\varepsilon_r) A e^{\frac{-E}{RT_s}} \tag{7}$$

Char conversion with O_2, H_2O and CO_2 is described with the shrinking core model. The bulk density of the char bed is constant, while the particle size and solid velocity are reduced during conversion. The volume of the particle (V_p) is reduced as a function of the char conversion (X_{char}), as show in Equation (8), and the solid velocity is reduced in a proportional way as shown in Equation (9), so that the bulk density is not modified. The reaction rate of char conversion for these reactions includes surface kinetics and the external mass transfer limitation due to diffusion of the reactant to the particle, as shown in Equation (10). A_p is the specific particle surface area and for spherical particles it is calculated as follows: $A_p = 6(1 - \varepsilon_r)/d_p$. The surface kinetics from Di Blasi [13] are employed, as shown in Table 1, as well as the mass transfer coefficient β. The CO/CO_2 ratio (η) in char oxidation from [30] is employed, as shown in Equation (11), which is derived from single particle combustion experiments with biomass char samples.

$$V_p = V_{p,0}(1 - X_{char}) \tag{8}$$

$$v_s = v_{s,0}\frac{V_p}{V_{p,0}} \tag{9}$$

$$\dot{r}_{char} = A_p M_{m,char} \frac{1}{\frac{1}{A e^{\frac{-E}{RT_s}}} + \frac{1}{\beta}} c_g X_{H_2O/CO_2/O_2} \tag{10}$$

$$\eta = \frac{12 e^{\frac{-3300}{T_s}}}{1 + 12 e^{\frac{-3300}{T_s}}} \tag{11}$$

Gas phase homogeneous reactions as gas phase oxidations or tar cracking are not included in the model. Gas phase oxidation is not included because the oxygen introduced at the bottom reacts very fast with the char in this zone and the obtained CO/CO_2 ratio of this heterogeneous reaction was already validated from single particle experiments [30]. Tar cracking is not included as the pyrolysis volatiles are released to the top and their temperatures are lower than 500 °C, which is needed for tar cracking [31]. The assumption of not including gas phase reactions will be discussed in the results section. The enthalpy of the reactions is calculated as the difference in heating value (Δh) of the reactants and products, and it is transferred to the solid phase (see Equation (4)).

Other model properties including reactor dimensions are shown in Table 2. The dry biomass particle density corresponds to spruce [32] and it is assumed that the particles are spherical with a diameter of 1 cm. The bed permeability is based on the assumed particle size and bed porosity [33]. The heat capacities of the permanent gas species are obtained from correlations based on values from the NIST database [34] between 300 and 1500 K. The heat of combustion of biomass, char and tar is based on the RAC scheme [26,27]. Other gas and solid properties are taken from Gronli [25].

Table 2
Model properties.

Property	Value	Units
Reactor length	0.25	m
Reactor diameter	0.25	m
Reactor porosity (ε_r)	0.6	–
Density dry biomass particle	430	$kg\cdot m^{-3}$
Initial particle diameter	0.01	m
Bed permeability (κ_s)	1×10^{-8}	m^2
Gas viscosity (μ_g)	3×10^{-5}	$kg\cdot m^{-1}\cdot s^{-1}$
Thermal conductivity biomass ($\lambda_{biomass}$)	0.17	$W\cdot m^{-1}\cdot K^{-1}$
Thermal conductivity char (λ_{char})	0.1	$W\cdot m^{-1}\cdot K^{-1}$
Thermal conductivity gas (λ_g)	0.0258	$W\cdot m^{-1}\cdot K^{-1}$
Solid emissivity	0.9	–
$\Delta h_{dry\ biomass}$	19.13	$MJ\cdot kg^{-1}$
Δh_{char}	32.37	$MJ\cdot kg^{-1}$
Δh_{tar}	21.44	$MJ\cdot kg^{-1}$
$c_{p,dry\ biomass}$	1500 + 1.0 T	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,char}$	420 + 2.09 T – 0.000685 T^2	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,moisture}$	4200	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,CO}$	979.7 + 0.193 T	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,CO2}$	594.3 + 0.977 T – 0.000331 T^2	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,H2O}$	1648 + 0.64 T	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,H2}$	14346–0.2679 T + 0.000917 T^2	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,CH4}$	1327 + 3.144 T	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,C2H4}$	238.2 + 4.854 T – 0.00176 T^2	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,tar}$	−100 + 4.4 T – 0.00157 T^2	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,O2}$	807 + 0.399 T – 0.000117 T^2	$J\cdot kg^{-1}\cdot K^{-1}$
$c_{p,N2}$	976.4 + 0.183 T	$J\cdot kg^{-1}\cdot K^{-1}$

The thermal conductivity of the bed (λ_s) is calculated according to Tsotas [35,36], including the conductivity without flow and the effect of flow through the Peclet number. Solid to gas heat transfer ($\dot{Q}_{s,g}$) is modelled according to Hobbs et al. [14], without the need for a correction factor as employed in the gasifier model by Di Blasi [13]. Solid ($\dot{Q}_{s,l}$) or gas ($\dot{Q}_{g,l}$) lateral heat transfer is calculated based on the heat transfer coefficient employed by Hobbs et al. [14] and the thermal resistance of the insulation. The insulation is surrounded by flowing water, so the heat transfer at this surface is not limiting the lateral heat transfer.

The reactor is discretized with 78 control volumes along the height, including a refining at the char conversion front and pyrolysis conversion front. It was checked that grid independent results are obtained with tests employing a higher number of control volumes. Convective terms are discretized with the upwind scheme and conductive terms with the central difference scheme [37]. The obtained system of ordinary differential equations is solved until convergence is achieved with the ode15s routine from Matlab, which is a multi-step solver based on numerical differentiation formulas and can handle stiff equations.

3. Results and discussion

3.1. Simulated cases

The simulated cases are shown in Table 3 and correspond to cases which have been experimentally investigated by Archan et al. [6]. Experiments were conducted in a small-scale biomass boiler which was designed to have a high fuel flexibility and to reduce emissions. The fuel conversion takes place in a fixed-bed with a low oxygen concentration, obtained through a low primary air ratio and the possibility to introduce flue gas recirculation. The input gases to the bed (primary air and eventually recirculation) are introduced to the bottom of the bed, under the grate, and biomass is introduced to the top of the bed in a counter-current configuration. The gases released from the bed are combusted in a secondary and tertiary zone located above the bed, which radiate energy to the bed. It has been simulated in this work with the previously described 1D reactor model the conversion in the bed of wood chips without flue gas recirculation with three different moisture contents (named as M8, M16 and M30) and with flue gas recirculation (reci) for two different moisture contents (named as M8 – with reci and M16 – with reci). The recirculated flue gas is the one obtained after a full oxidation of the gases in the combustion chamber above the fixed-bed, not the one released from the fixed-bed.

In the experimental work there were slight variations in biomass input power and air equivalent ratios for each case, as it is challenging to establish very precisely the biomass flow input. For simplicity an averaged biomass input power of 30.3 kW is selected for all cases. The amount of primary air was selected in the cases without flue gas recirculation to obtain an equivalent ratio of 0.213, which is the average obtained experimentally. With flue gas recirculation, the amount of primary air leads to an equivalent ratio of 0.176, which is as well the average obtained experimentally. The real equivalent ratio including the flue gas recirculation is however of around 0.22. The composition in mass percentage of the flue gas recirculation, obtained from a full combustion, is as follows: 6.4% O_2, 18.9% CO_2, 6.8% H_2O, 67.9% N_2 for M8 – with reci and 6.4% O_2, 18.6% CO_2, 7.8% H_2O, 67.2% N_2 for M16 – with reci.

The values employed for the heat losses from bottom, heat input from the top and inlet gas temperature are based on a detailed CFD simulation for the case M8 (results not shown). The lateral heat release is calculated with a thermal resistance of the insulation of $1/15\ K\cdot m^2\cdot W^{-1}$. It should be considered that most of the heat released from the bed to the surroundings (lateral and bottom) is mainly transferred to the water flowing in the boiler, so it is not a heat loss when considering the whole boiler system. Besides, around the half of heat losses through the bottom are employed in pre-heating the inlet gases (primary air and flue gas recirculation) in the reactor, which is also not a real loss. For other cases than M8 the same boundary conditions are employed for heat losses and input, including the energy employed in pre-heating the inlet air, except the cases with flue gas recirculation where the heat losses through the bottom are lower due to lower grate temperatures. The employed values will be later on discussed. Furthermore, it is assumed that during the experimental measurements in [6], which were used for model validation, an ash layer of approximately 0.01 m was present at the bottom of the reactor, which is not modeled. A temporal accumulation of ash particles takes place during the experiments on the grate forming an ash layer. The assumed ash layer height was calculated from the typical operating time of the reactor, the grate geometry and measured ash density and amount of ash supplied via the fuel.

Table 3
Boundary conditions for simulated cases.

	Unit	M8	M16	M30	M8 – with reci	M16 – with reci
Biomass input	$kg\cdot h^{-1}$	6.25	7	8.7	6.25	7
Moisture content	% mass w.b.	8	16.5	30.5	8	16.5
Primary air	$kg\cdot h^{-1}$	7.4	7.5	7.75	6.1	6.2
Flue gas recirculation	$kg\cdot h^{-1}$	–	–	–	5.1	5.7
Inlet gas temperature	°C	500	500	500	350	325
Heat loss bottom	kW	2.0	2.0	2.0	1.8	1.8
Heat input top	kW	0.75	0.75	0.75	0.75	0.75

3.2. Analysis of conversion for case M30

The model results are first presented in detail for the case M30. The solid and most relevant gas species mass flows as well as temperatures along the reactor height are shown in Fig. 1 and the gas composition along the reactor height in Fig. 2.

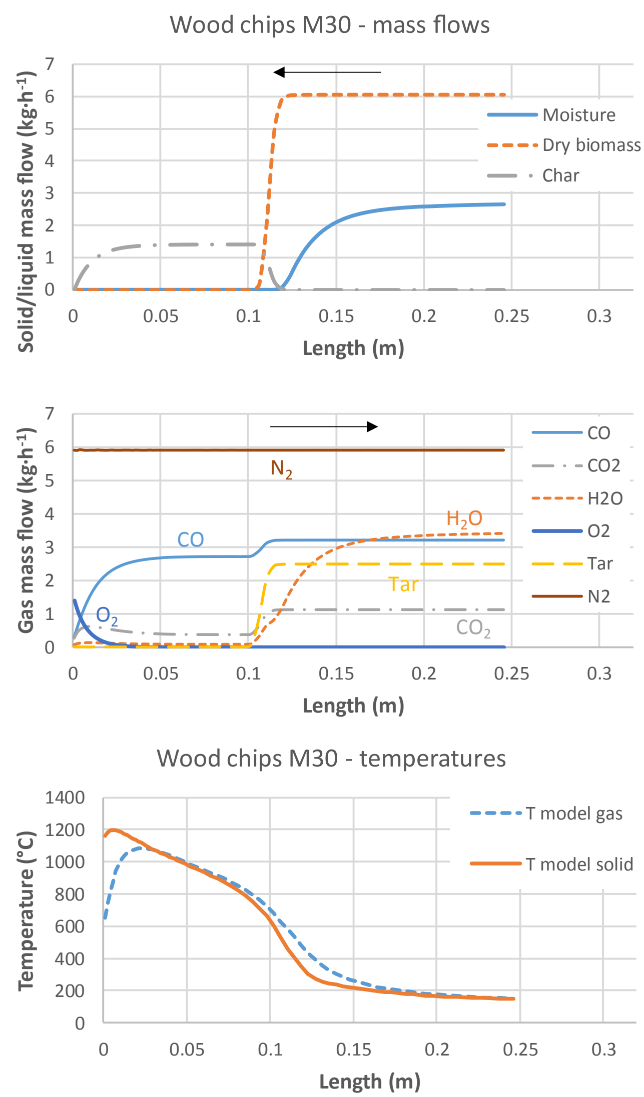

Fig. 1. Mass flow of solid and liquid (top) and most relevant gas (middle) species as well as solid and gas temperatures (bottom) along the reactor height for the case wood chips M30.

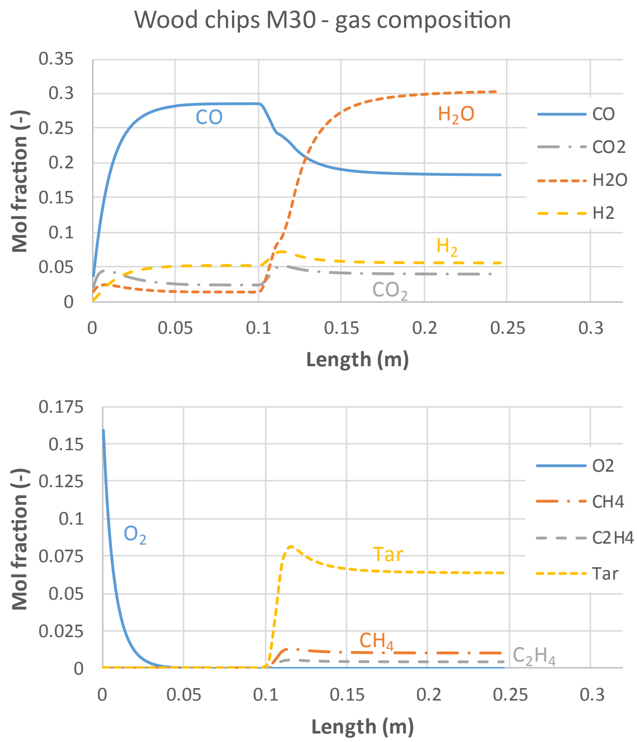

Fig. 2. Gas composition along the reactor height for the case wood chips M30.

Primary air is introduced at the bottom of the reactor, and oxygen is already consumed after a few cm. The highest temperature is achieved at this position, where char is oxidized. Besides, char reduction reactions take place as well and the main gas product of the char conversion zone is CO, followed by H_2, CO_2 and H_2O. The pyrolysis front is placed at around 12 cm from the bottom and at this stage a significant amount of tars is generated, as well as to a lower extent of other gases as CO_2, CO, CH_4, C_2H_4, H_2O and H_2. Above it drying takes place, generating more H_2O and reducing the fixed-bed temperatures, achieving the minimum at the top of the reactor. This behavior resembles a typical updraft biomass gasifier [13].

The mass and energy balances for the M30 case are shown in Table 4. The char conversion is almost complete at these conditions. Therefore, biomass and primary air input mass flows are mainly converted into the producer gas. The error in the mass balance is minimal.

The biomass input energy (30.3 kW) is mainly transferred to the producer gas (28.6 kW chemical energy, 0.8 kW thermal energy). The producer gas composition is detailed in Table 5 and it is characterized by a high tar content. The error in the energy balance is minor. The lateral heat losses are of 1.0 kW and the remaining energy in the output char is only 0.2 kW. The heat radiation to the top (0.8 kW) and heat losses to the bottom (2 kW) were model assumptions, as well as the considered inlet primary air temperature, which leads to a pre-heating of 1.1 kW which are obtained from the heat losses to the bottom. The total heat losses of the reactor, considering inputs and outputs, are of 1.2 kW. However, a significant fraction is transferred to the water flowing in the boiler.

Table 4
Mass and energy balances for case M30.

	Mass (kg·s^{-1})		Power (kW)
Inputs			
Biomass	8.70	Biomass - combustion enthalpy	30.3
Primary air	7.75	Primary air - latent enthalpy	1.1
–	–	Heat input top	0.8
Outputs			
Producer gas	16.41	Producer gas - combustion enthalpy	28.5
–	–	Producer gas - latent enthalpy	0.7
Char	0.02	Char - combustion enthalpy	0.2
–	–	Lateral heat losses	1.0
–	–	Heat loss bottom	2.0

Table 5
Producer gas composition (wet basis) for the case M30 comparing experiments [6] and model.

	Measured	Modeled	Difference
CO (% mass)	18.2	19.6	+1.4
CO_2 (% mass)	7.3	6.9	−0.4
H_2O (% mass)	22.4	20.9	−1.5
H_2 (% mass)	0.3	0.4	+0.1
CH_4 (% mass)	0.7	0.6	−0.1
C_2H_4 + other C_xH_y (% mass)	0.4	0.5	+0.1
Tar (% mass)	14.4	15.1	+0.7

3.3. Model validation for all cases

The comparison between experiments and model results for temperatures in the reactor is shown in Fig. 3. A generally good agreement is achieved for all cases. Temperatures were experimentally measured at 3 positions inside the reactor and the model can correctly predict the temperature profiles for all cases. The experimental standard deviations along the height consider the possible variations in the exact position of the thermocouples. The highest temperatures are achieved at the bottom of the reactor, close to the grate. The achieved maximum temperatures are shown in Table 6. The temperatures at the top of the reactor are reduced with a higher moisture content due to drying, which takes place at this position. A higher moisture content moves the drying and pyrolysis fronts to lower heights, as more time is needed for the solid particles to achieve the required temperatures for these processes. However, the maximum temperature in the reactor is not affected by the fuel moisture content. This could be explained because, even for the maximum moisture content (M30 – see Fig. 1), the pyrolysis and drying fronts are still at a significant distance from the char conversion front. Therefore, drying taking place to a higher or lower extent is not affecting the char conversion zone.

A reduction of the maximum solid temperature of the reactor was however achieved with flue gas recirculation, obtaining a peak temperature which is about 130 °C lower than without flue gas recirculation. The obtained reduction in temperature was correctly simulated by the model, as the difference between the maximum solid temperature in the model and the experimentally measured temperature at the lowest position (which is not exactly at the same position as the maximum in the model) was approximately 100 °C for all cases (see Table 6). This was therefore the only measure that could be employed to reduce temperatures on the grate with this technology, which would be needed to avoid slagging with complicated fuels.

Temperatures were also measured above the bed (0.31 m above the grate). At this position a direct comparison with the model is not possible, as only the fixed-bed is modeled, with a height of 0.25 m. Besides, these temperature measurements are affected by the radiation from the top of the reactor. However, it is also well predicted by the model the trend that temperatures of the producer gas are reduced with a higher moisture content.

Regarding producer gas composition, a good agreement between experiments and model results is found for all compounds, as shown in Fig. 4. This shows the suitability of the proposed reaction scheme in Section 2, including the product composition of the pyrolysis reaction which is based on a detailed scheme. As the reactor model is 1D, the spatial distribution of the volatiles release which has been experimentally measured [6] cannot be described, so the comparison with the model is done with the average producer gas composition.

The CO and CO_2 yields are well predicted for all cases. There is a very

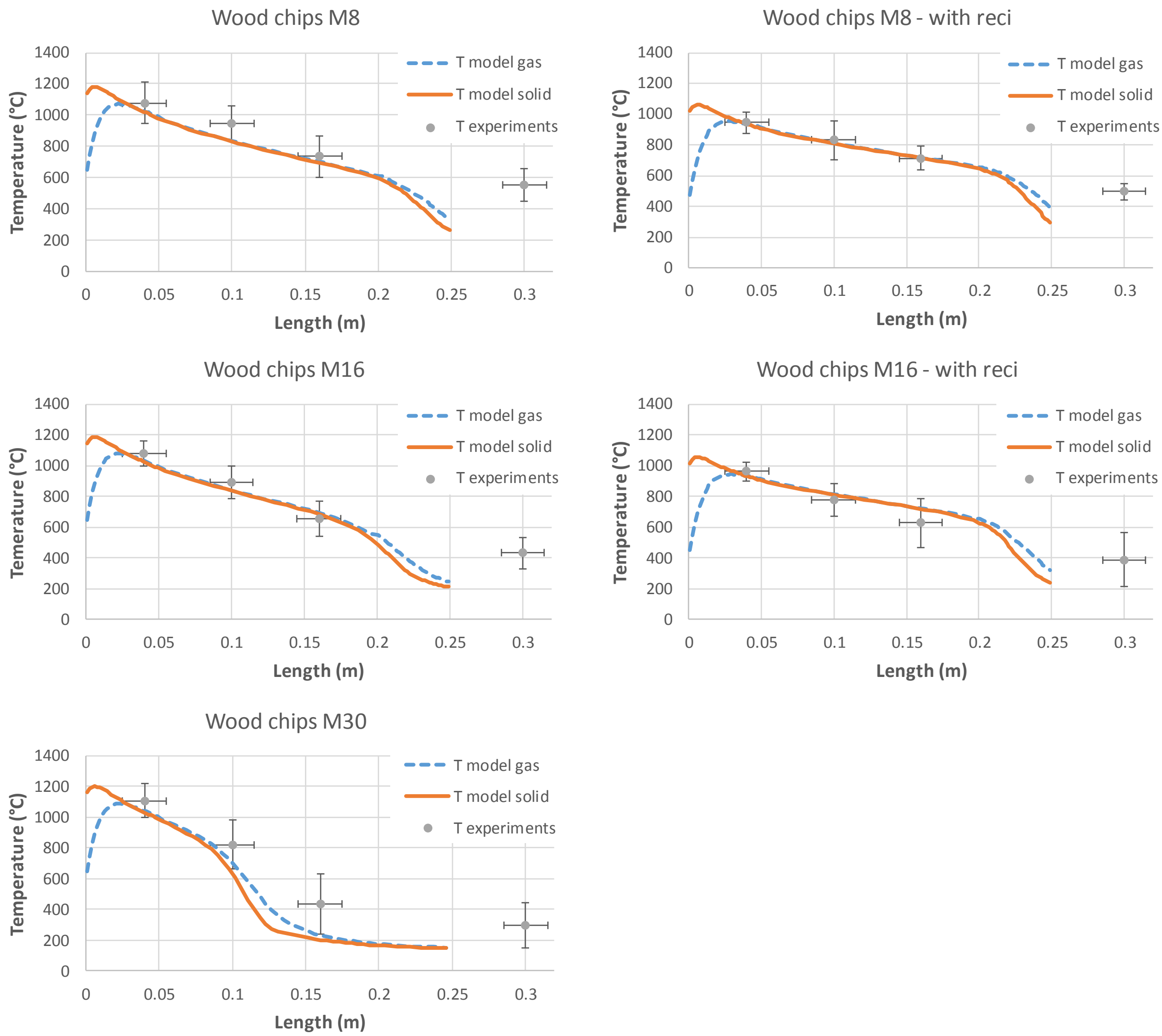

Fig. 3. Temperatures along the reactor height for all cases comparing experiments [6] and model (assumed that during the experiments a 0.01 m ash layer was present at the bottom of the reactor, which is not modeled).

Table 6
Maximum modeled solid temperatures, measured temperatures at 0.05 m above the grate [6] and difference between both values (which do not correspond to the same height).

Case	Maximum solid temperature model (°C)	Measured temperature at 5 cm above grate (°C)	Difference (°C)
M8	1179	1079	−100
M16	1184	1080	−104
M30	1197	1106	−90
M8 – with reci	1060	947	−112
M16 – with reci	1054	960	−94

high CO yield, specially without flue gas recirculation. This arises from the efficient char reduction zone, which is therefore correctly simulated. The differences in CO and CO_2 yields between model and experiments are only noticeable for the case M16 (without recirculation). However, in this case the experimental equivalent air ratio was slightly higher (0.23) than for other cases, which was not considered in the model and probably lead to a lower CO/CO_2 ratio in the experiments. It has to be considered that the experiments were conducted with a reactor which is close to commercialization, and the 1D homogeneity is not perfect.

The H_2O yield is slightly underpredicted for all cases, and the H_2 yield over-predicted (see also Table 5). The H_2 yield is low in mass fraction but around 5% in wet volume fraction. The content of total light hydrocarbons is well predicted, including CH_4, C_2H_4 and other light hydrocarbons. In the model only CH_4 and C_2H_4 are considered. C_2H_4 in the model is also representative for other light hydrocarbons (see also Table 5).

The total tar content is very well predicted and the modeled tar contents are only slightly higher than the ones experimentally measured for all cases. Tar in the model represents all condensable species, including the gravimetric tar and light condensable species experimentally measured. These results show that it is not needed to described tar cracking in bed for this technology. Furthermore, for the case M8, with the highest temperatures at the pyrolysis front, a simulation was conducted including the tar cracking kinetics from Liden et al. [38] ($A = 4.26 \times 10^6\ s^{-1}$, $E = 108\ kJ{\cdot}mol^{-1}$), which are the fastest among the most commonly employed tar cracking kinetics in literature [31]. The results showed that less than 0.2% of the tar was cracked for this case, confirming that tar cracking in the bed is minimal at these conditions. Tar cracking will then take place above the bed, in the secondary zone of the boiler.

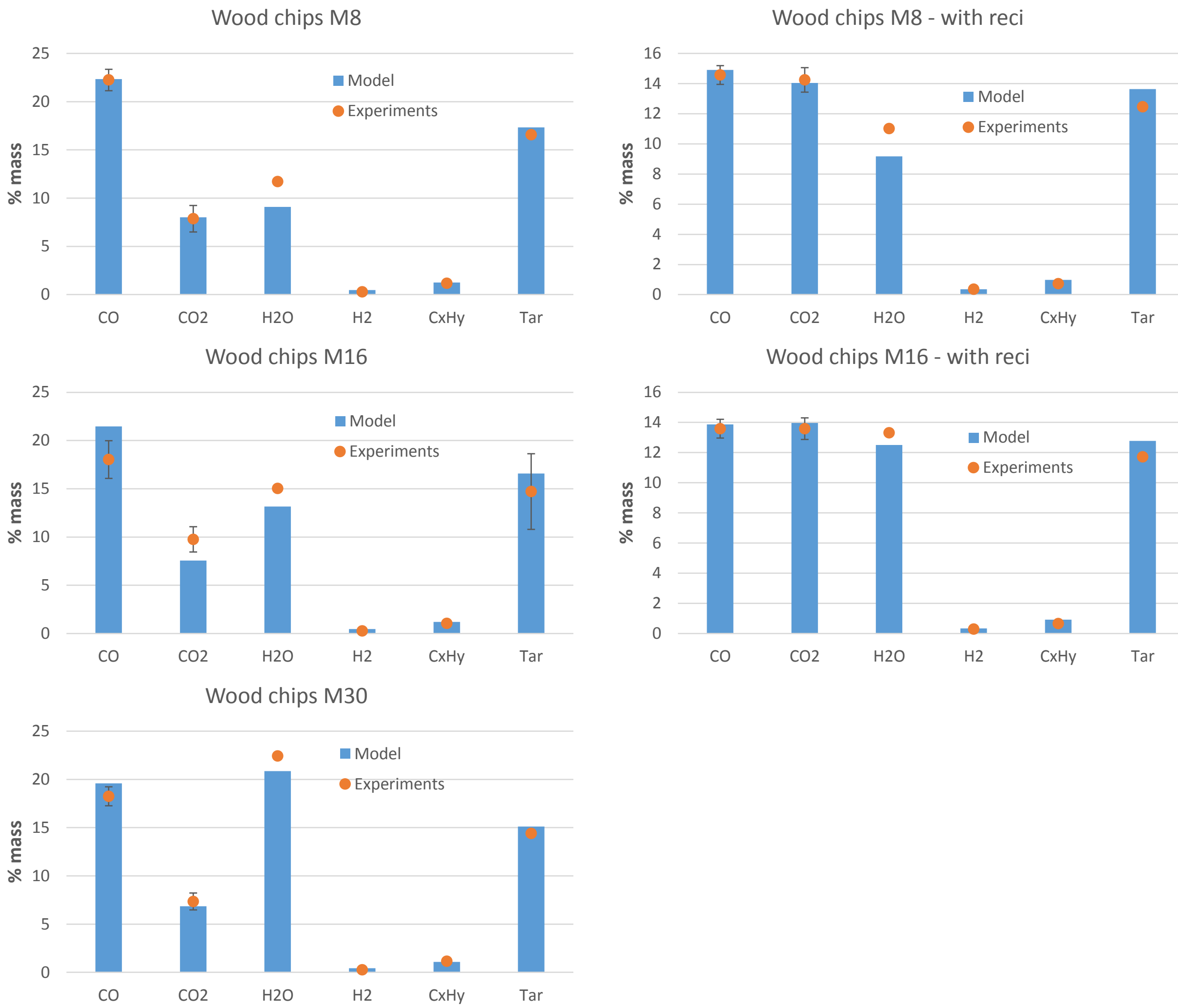

Fig. 4. Producer gas composition for all cases comparing experiments [6] (including standard deviations which are available for permanent gases and for tar in case M16) and model (C_xH_y represents the sum of all light hydrocarbons, including CH_4 and C_2H_4).

For additional validation, the producer gas composition at the middle of the reactor was measured for the case M30 and compared to the model results. The measurement equipment is the same as in Archan et al. [6]. The comparison between measured and modelled compositions is shown in Table 7. There is a good agreement for most of the compounds, except for CH_4. The model correctly predicts that the main species is CO, obtaining at this stage a low CO_2 yield due to an effective reduction process. Besides, the tar content is minimal at this position. Looking at Fig. 1 and Fig. 2, it can be seen that the model predicts that the pyrolysis front ends just above this measurement point. The discrepancy in the CH_4 content can be attributed to a late CH_4 release that commonly takes place in pyrolysis, after the main peaks of release of condensable compounds and main gases, as already detected in slow and intermediate pyrolysis experiments with single particles [26,39].

Table 7
Gas composition at the middle of the reactor for the case M30 (0.11 m above the grate – 0.10 m reactor height in the model due to the assumption of a 0.01 m ash layer at the bottom).

	Measured	Modeled
CO (% vol. dry)	30.3 ± 2.3	29.0
CO_2 (% vol. dry)	4.2 ± 0.7	2.6
H_2 (% vol. dry)	6.6 ± 2.9	5.3
CH_4 (% vol. dry)	2.0 ± 1.3	0.03
Tar (% vol. dry)	≈ 0	0.2

Therefore, the most probable explanation is that the position of the main pyrolysis front (just above this measurement point) is well described by the model for this case, and what is detected is CH_4 from pyrolysis which is not modelled as pyrolysis is described with only one reaction, while CH_4 in pyrolysis is released after the main compounds. Besides, intraparticle gradients are not described in the model, which can delay the pyrolysis process.

3.4. *Discussion of the results*

The presented model was successfully validated. This shows that the model assumptions were reasonable for this case. This includes not considering gas phase reactions as oxidations or water gas shift, together with the employed CO/CO_2 ratio. The current model already leads to good predictions for this technology of the temperatures, including temperatures close to the grate, and gas composition at the middle and top of the reactor, correctly predicting the char reduction process producing a high CO yield. Tar cracking was also not required to describe this process. Besides, the gas and solid phases were separated in the energy equation [40] and a correction of the gas to solid heat transfer description proposed by Hobbs et al. [14] was not needed, as done in other literature works [13].

The modeled technology is similar to conventional updraft gasifiers, although there are some differences. Updraft gasifiers are commonly longer, while in this case the height is relatively short and besides there is a heat input to the top by radiation from the combustion chamber. The most relevant difference in the producer gas composition in comparison to literature is the tar content. Literature reports typical tar contents for updraft gasifiers (30 – 150 g/Nm^3 [41], 20–100 g/Nm^3 [42]) lower than the experimentally detected and modelled in this work (>200 g/Nm^3 [6]). It has to be stated that in several gasifiers it is actually targeted to reduce the tar content for several applications of the gas, while here it is not a problem as the producer gas is directly burnt above the fixed-bed. It has been shown in this work that even for the fuel with a low moisture content, the achieved temperatures were not sufficient so that homogeneous tar cracking reactions in the reactor were relevant. It was suggested by Cerone et al. [22] that tar condensation inside the bed and decomposition was a main mechanism to reduce the tar content in updraft gasification. Condensation of tars can take place to a significant extent at temperatures around 200 °C or lower [43]. In this work temperatures in these range were only achieved for the case M30, but the experimental results showed that it was not enough to achieve a reduction in the tar content. In conventional updraft gasifiers which are longer this will be more typical due to absence of heat radiation to the top, lower temperatures at the top and middle of the gasifier due to a bigger separation between the exothermic char conversion zone and pyrolysis/drying zone and the longer residence time of the volatiles in the fixed-bed.

This work provides relevant inputs to develop combustion technologies with a low oxygen concentration in the fixed-bed. The results have shown that there is a clear separation of the char conversion zone and the pyrolysis/drying zone. Variations in the moisture content between 8 and 30% w.b. lead to displacements of the pyrolysis and drying fronts, but do not influence the producer gas composition or the maximum temperatures achieved in the grate. Despite the relatively compact design a significant char reduction producing a high CO yield was achieved for all cases, although for the wet fuel this zone is reduced. This shows the flexibility of the technology regarding the fuel moisture content. Besides, this could have a significant impact regarding NO_X emissions, as the NO that should be obtained from nitrogen in the char during its oxidation at the bottom may be reduced as well to N_2 due to the catalytic activity of char in this zone [44]. Furthermore, for fuels with a low ash melting point, primary flue gas recirculation is the only possibility to reduce grate temperatures avoiding slagging on the grate with this reactor configuration. This was experimentally shown with Miscanthus in [6]. The model presented in this work allows the calculation of the reduction of maximum temperatures which is achieved with a certain amount of flue gas recirculation.

The application of this model can provide boundary conditions regarding producer gas composition and temperature for detailed CFD simulations of the gas phase combustion in the boiler. However, the currently presented model is 1D, which is a simplification of a more complex process. CFD simulations considering the movement of solid particles, as currently done e.g with moving grates [45], would be required for a more detailed prediction of the bed conversion behavior with this technology. In this way it would be possible to directly calculate the heat losses and inputs from bottom and top of the reactor or reactor height (which are inputs of the current 1D model) for different geometries, and to investigate in detail their influence on the conversion behavior. Moreover, a detailed 3D CFD model would also be needed to describe the spatial distribution of the release of volatiles, which does not have to be homogeneous along the cross section of the reactor, as assumed in the 1D model. To develop such a CFD simulation is a big challenge for future work, and this work establishes the basis of the main conversion behavior that should be also accurately described by more detailed models.

4. Conclusions

The presented 1D model can describe updraft gasification of biomass and it was employed to model the bed conversion in a combustion system with a low primary air ratio and a counter-current configuration. The model can predict with a good accuracy the temperature evolution in the reactor and producer gas composition of different cases with variations in fuel moisture content and inclusion in some of them of flue gas recirculation. This shows the suitability of the proposed reaction scheme, including the product composition of the pyrolysis reaction which is based on a detailed scheme. The 1D model shows that the char conversion and pyrolysis/drying zones are well separated, despite the limited height of the reactor. Higher moisture contents reduce the temperatures at the top of the reactor but do not have any influence in the char conversion zone for the tested moisture contents (8 to 30% mass w.b.), e.g. regarding maximum temperatures and presence of reducing conditions mainly obtaining CO. Also the dry producer gas composition is not affected by the fuel moisture content and higher tar contents than in conventional updraft gasification are achieved, probably because tars do not condense in the fixed-bed due to the short height of the reactor and the heat input from radiation of the combustion chamber. Flue gas recirculation is the only possibility to reduce the maximum bed temperatures in the selected reactor configuration, enabling the use of fuels with a low ash melting point. The influence of the flue gas recirculation was correctly predicted by the simulations. The presented model is an efficient tool to support the development of this flexible combustion technology. This is done through advanced process analysis as well as providing boundary conditions for detailed CFD simulations of the gas phase combustion or being the basis for more detailed 3D CFD models of the bed conversion including particle movement.

CRediT authorship contribution statement

Andrés Anca-Couce: Conceptualization, Formal analysis, Writing - original draft. **Georg Archan:** Validation, Writing - review & editing. **Markus Buchmayr:** Resources. **Michael Essl:** Software. **Christoph Hochenauer:** Resources. **Robert Scharler:** Validation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Anca-Couce A, Hochenauer C, Scharler R. Bioenergy technologies, uses, market and future trends with Austria as a case study. Renew Sustain Energy Rev 2021;135: 110237. https://doi.org/10.1016/j.rser.2020.110237.
[2] Feldmeier S, Wopienka E, Schwarz M, Schön C, Pfeifer C. Applicability of Fuel Indexes for Small-Scale Biomass Combustion Technologies, Part 2: TSP and NOx Emissions. Energy Fuels 2019;33(11):11724–30.
[3] Sommersacher P, Brunner T, Obernberger I. Fuel indexes: a novel method for the evaluation of relevant combustion properties of new biomass fuels. Energy Fuels 2012;26(1):380–90.
[4] Feldmeier S, Wopienka E, Schwarz M, Schön C, Pfeifer C. Applicability of Fuel Indexes for Small-Scale Biomass Combustion Technologies, Part 1: Slag Formation. Energy Fuels 2019;33(11):10969–77.
[5] Nussbaumer T. Combustion and co-combustion of biomass: fundamentals, technologies, and primary measures for emission reduction. Energy Fuels 2003;17 (6):1510–21.
[6] Archan G, Anca-Couce A, Gregorc J, Buchmayr M, Hochenauer C, Gruber J, et al. Detailed experimental investigation of the spatially distributed gas release and bed temperatures in fixed-bed biomass combustion with low oxygen concentration.

Biomass Bioenergy 2020;141:105725. https://doi.org/10.1016/j.biombioe.2020.105725.
[7] Buchmayr M, Gruber J, Hargassner M, Hochenauer C. Experimental investigation of the primary combustion zone during staged combustion of wood-chips in a commercial small-scale boiler. Biomass Bioenergy 2015;81:356–63.
[8] Obernberger I, Brunner T, Mandl C, Kerschbaum M, Svetlik T. Strategies and technologies towards zero emission biomass combustion by primary measures. Energy Procedia 2017;120:681–8.
[9] Porteiro J, Patiño D, Miguez JL, Granada E, Moran J, Collazo J. Study of the reaction front thickness in a counter-current fixed-bed combustor of a pelletised biomass. Combust Flame 2012;159(3):1296–302.
[10] Molino A, Chianese S, Musmarra D. Biomass gasification technology: The state of the art overview. Journal of Energy Chemistry 2016;25(1):10–25.
[11] Sikarwar VS, Zhao M, Fennell PS, Shah N, Anthony EJ. Progress in biofuel production from gasification. Prog Energy Combust Sci 2017;61:189–248.
[12] Olwa J, Öhman M, Esbjörn P, Boström D, Okure M, Kjellström B. Potassium retention in updraft gasification of wood. Energy Fuels 2013;27(11):6718–24.
[13] Di Blasi C. Modeling wood gasification in a countercurrent fixed-bed reactor. AIChE J 2004;50(9):2306–19.
[14] Hobbs ML, Radulovic PT, Smoot LD. Modeling fixed-bed coal gasifiers. AIChE J 1992;38(5):681–702.
[15] Mandl C, Obernberger I, Biedermann F. Modelling of an updraft fixed-bed gasifier operated with softwood pellets. Fuel 2010;89(12):3795–806.
[16] Yang W, Ponzio A, Lucas C, Blasiak W. Performance analysis of a fixed-bed biomass gasifier using high-temperature air. Fuel Process Technol 2006;87(3):235–45.
[17] Umeki K, Namioka T, Yoshikawa K. Analysis of an updraft biomass gasifier with high temperature steam using a numerical model. Appl Energy 2012;90(1):38–45.
[18] Zhang Q, Dor L, Biswas AK, Yang W, Blasiak W. Modeling of steam plasma gasification for municipal solid waste. Fuel Process Technol 2013;106:546–54.
[19] Di Blasi C, Branca C. Modeling a stratified downdraft wood gasifier with primary and secondary air entry. Fuel 2013;104:847–60.
[20] Yucel O, Hastaoglu MA. Kinetic modeling and simulation of throated downdraft gasifier. Fuel Process Technol 2016;144:145–54.
[21] Sommariva S, Grana R, Maffei T, Pierucci S, Ranzi E. A kinetic approach to the mathematical model of fixed bed gasifiers. Comput Chem Eng 2011;35(5):928–35.
[22] Cerone N, Zimbardi F, Contuzzi L, Baleta J, Cerinski D, Skvorčinskienė R. Experimental investigation of syngas composition variation along updraft fixed bed gasifier. Energy Convers Manage 2020;221:113116. https://doi.org/10.1016/j.enconman.2020.113116.
[23] Li B, Chen H, Yang H, Wang X, Zhang S. Characteristics of the temperature distribution and product gas evolving of an updraft biomass gasifier. Energy Fuels 2013;27(3):1460–5.
[24] Jakobsen HA, Lindborg H, Handeland V. A numerical study of the interactions between viscous flow, transport and kinetics in fixed bed reactors. Comput Chem Eng 2002;26(3):333–57.
[25] Grønli MG. A theoretical and experimental study of the thermal degradation of biomass. PhD thesis. Norwegian University of Science and Technology (NTNU); 1996.
[26] Anca-Couce A, Sommersacher P, Scharler R. Online experiments and modelling with a detailed reaction scheme of single particle biomass pyrolysis. J Anal Appl Pyrol 2017;127:411–25.
[27] Anca-Couce A, Scharler R. Modelling heat of reaction in biomass pyrolysis with detailed reaction schemes. Fuel 2017;206:572–9.
[28] Anca-Couce A, Obernberger I. Application of a detailed biomass pyrolysis kinetic scheme to hardwood and softwood torrefaction. Fuel 2016;167:158–67.
[29] Di Blasi C. Combustion and gasification rates of lignocellulosic chars. Prog Energy Combust Sci 2009;35(2):121–40.
[30] Anca-Couce A, Sommersacher P, Shiehnejadhesar A, Mehrabian R, Hochenauer C, Scharler R. CO/CO2 ratio in biomass char oxidation. Energy Procedia 2017;120:238–45.
[31] Gómez-Barea A, Leckner B. Modeling of biomass gasification in fluidized bed. Prog Energy Combust Sci 2010;36(4):444–509.
[32] Kaltschmitt M, Hartmann H, Hofbauer H. Energie aus Biomasse. Grundlagen: Techniken und Verfahren (in German), Springer-Verlag; 2010.
[33] Thess A. Heat Transfer by Free Convection: Internal Flows. VDI Heat Atlas. Second Edition (section F3), Springer-Verlag; 2010.
[34] Afeefy HY, Liebman JF, Stein SE. Neutral Thermochemical Data. In: Linstrom PJ, Mallard WG, editors. NIST Chemistry Webbook, NIST Standard Reference Database Number 69. Gaithersburg: MD; 2001.
[35] Tsotsas E. Heat and Mass Transfer in Packed Beds with Fluid Flow. VDI Heat Atlas. Second Edition (section M7), Springer-Verlag; 2010.
[36] Tsotsas E. Thermal Conductivity of Packed Beds. VDI Heat Atlas. Second Edition (section D6.3), Springer-Verlag; 2010.
[37] Anca-Couce A, Zobel N. Numerical analysis of a biomass pyrolysis particle model: Solution method optimized for the coupling to reactor models. Fuel 2012;97:80–8.
[38] LIDEN AG, BERRUTI F, SCOTT DS. A kinetic model for the production of liquids from the flash pyrolysis of biomass. Chem Eng Commun 1988;65(1):207–21.
[39] Almuina-Villar H, Lang N, Anca-Couce A, Röpcke J, Behrendt F, Dieguez-Alonso A. Application of laser-based diagnostics for characterization of the influence of inorganics on the slow pyrolysis of woody biomass. J Anal Appl Pyrol 2019;140:125–36.
[40] Cooper J, Hallett WLH. A numerical model for packed-bed combustion of char particles. Chem Eng Sci 2000;55(20):4451–60.
[41] Heidenreich S, Foscolo PU. New concepts in biomass gasification. Prog Energy Combust Sci 2015;46:72–95.
[42] La Villetta M, Costa M, Massarotti N. Modelling approaches to biomass gasification: A review with emphasis on the stoichiometric method. Renew Sustain Energy Rev 2017;74:71–88.
[43] Anca-Couce A, Brunner T, Kanzian W, Obernberger I, Trattner K. Characterization and condensation behaviour of gravimetric tars produced during spruce torrefaction. J Anal Appl Pyrol 2016;119:173–9.
[44] Karlström O, Brink A, Hupa M. Time dependent production of NO from combustion of large biomass char particles. Fuel 2013;103:524–32.
[45] Bermúdez CA, Porteiro J, Varela LG, Chapela S, Patiño D. Three-dimensional CFD simulation of a large-scale grate-fired biomass furnace. Fuel Process Technol 2020;198:106219. https://doi.org/10.1016/j.fuproc.2019.106219.